

INTELLIGENT PREDICTIVE MODELLING USING BIG DATA FOR GENERIC DRUG SELECTION

THESIS SUBMITTED TO
SAVITRIBAI PHULE PUNE UNIVERSITY

For

AWARD OF DEGREE OF

Doctor of Philosophy(Ph.D.)

In the
Faculty of Computer Management

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Progressive Education Society's
Modern Institute of Business Studies, Pune Maharashtra

Agenda



- Introduction
- Traditional Drug selection Process
- Literature Review
- Aim and Objectives
- Design
- Model Diagram
- Flowchart
- Data Collection
- Input->Process->Output
- Data Visualization
- Conclusion
- Publications
- Limitations
- Future Research
- References

Introduction

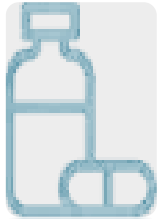


Stands 3rd world wide in production and 14th in terms of Value

Types of Drug

Innovator Drugs

Generic Drugs



Drug

DP(Formulation)

API

Drug Product(Formulation)

Drug Content



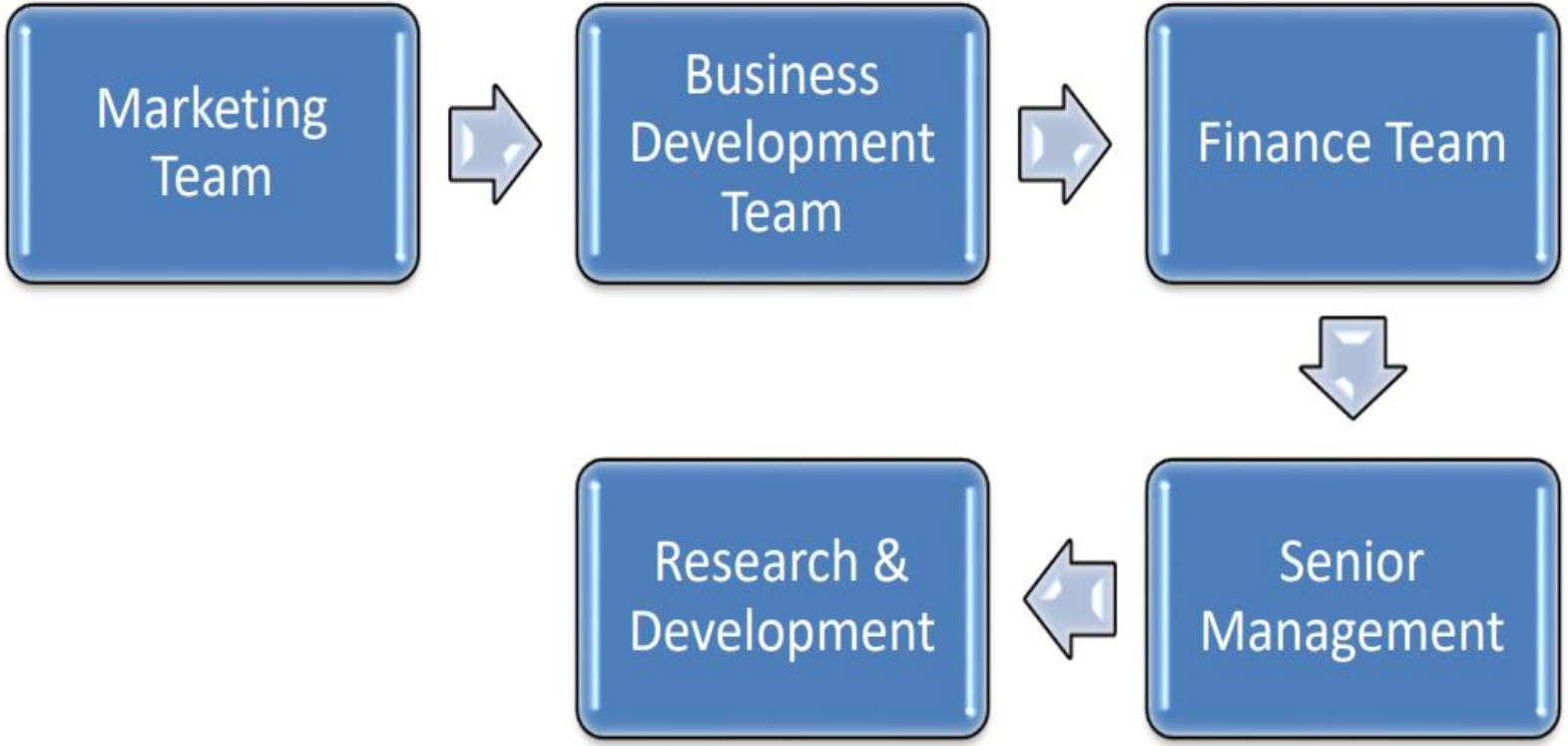
- ✓Tablets
- ✓Capsules
- ✓Injections(Vial)
- ✓Syrup
- ✓Liquid
- ✓Ointments
- ✓Powder
- ✓Gel



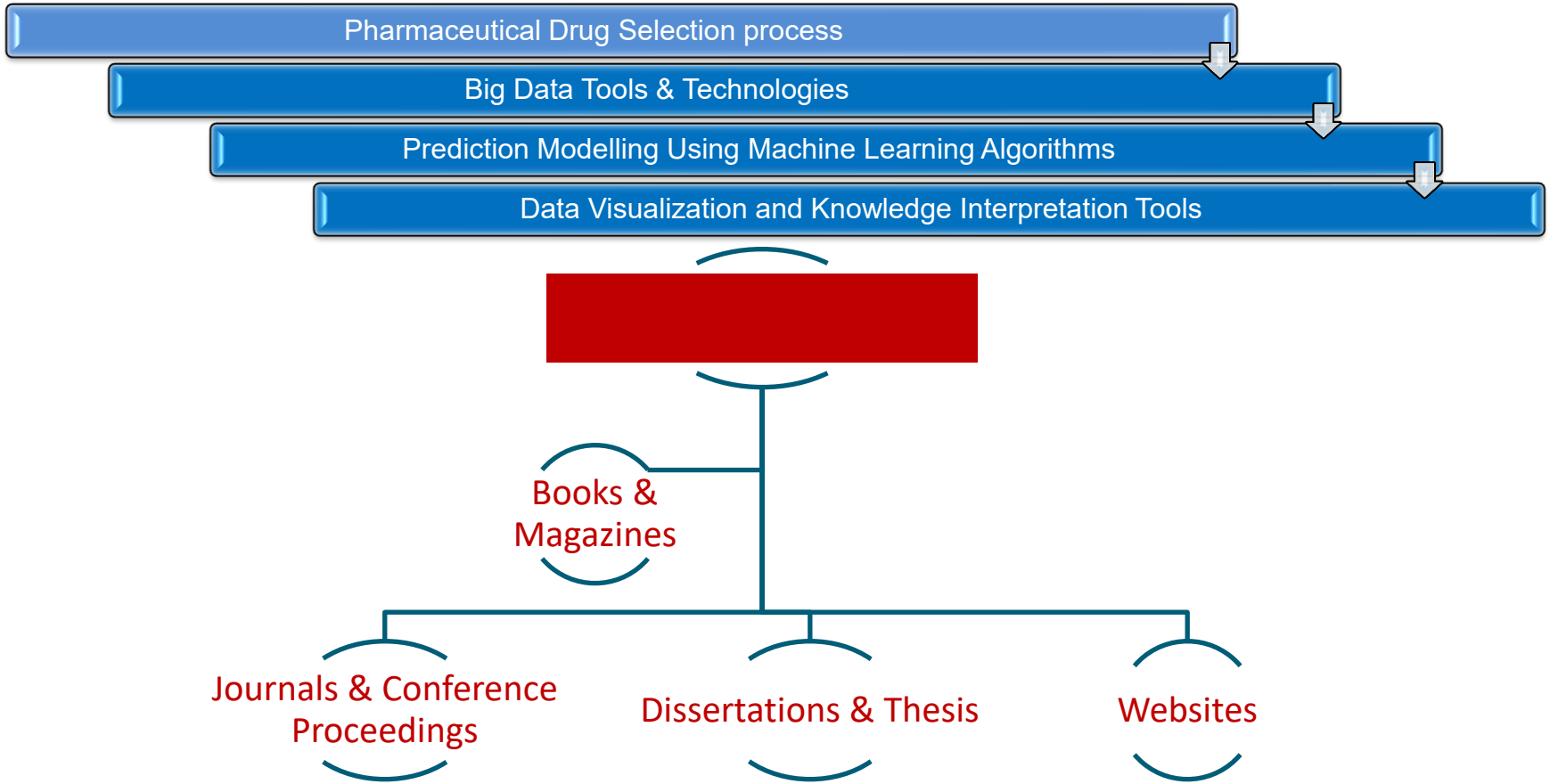
Active Pharmaceutical Ingredient

- ✓ Raw material for DP
- ✓ Variety of states, such as crystals and powders
- ✓ E.g. In Crocin tablet Paracetamol substance act as API
- ✓ API for relief of pain is included in painkiller

Traditional Drug Selection Process



Literature Review



Aim & Objectives

AIM

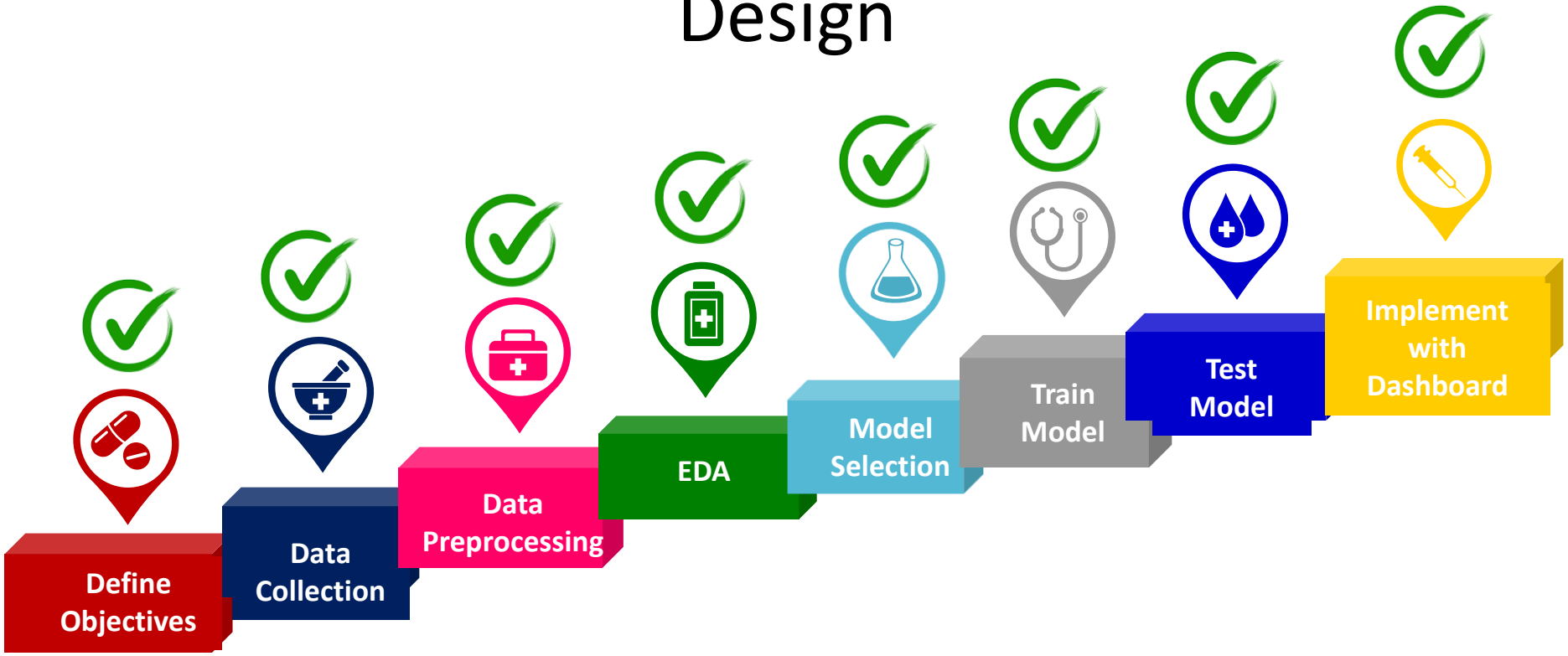
The primary aim of the proposed model is to utilize effectively big data analytical techniques and to develop an Intelligent Predictive model which will be acting as a guideline for the pharmaceutical product selection.

Objectives

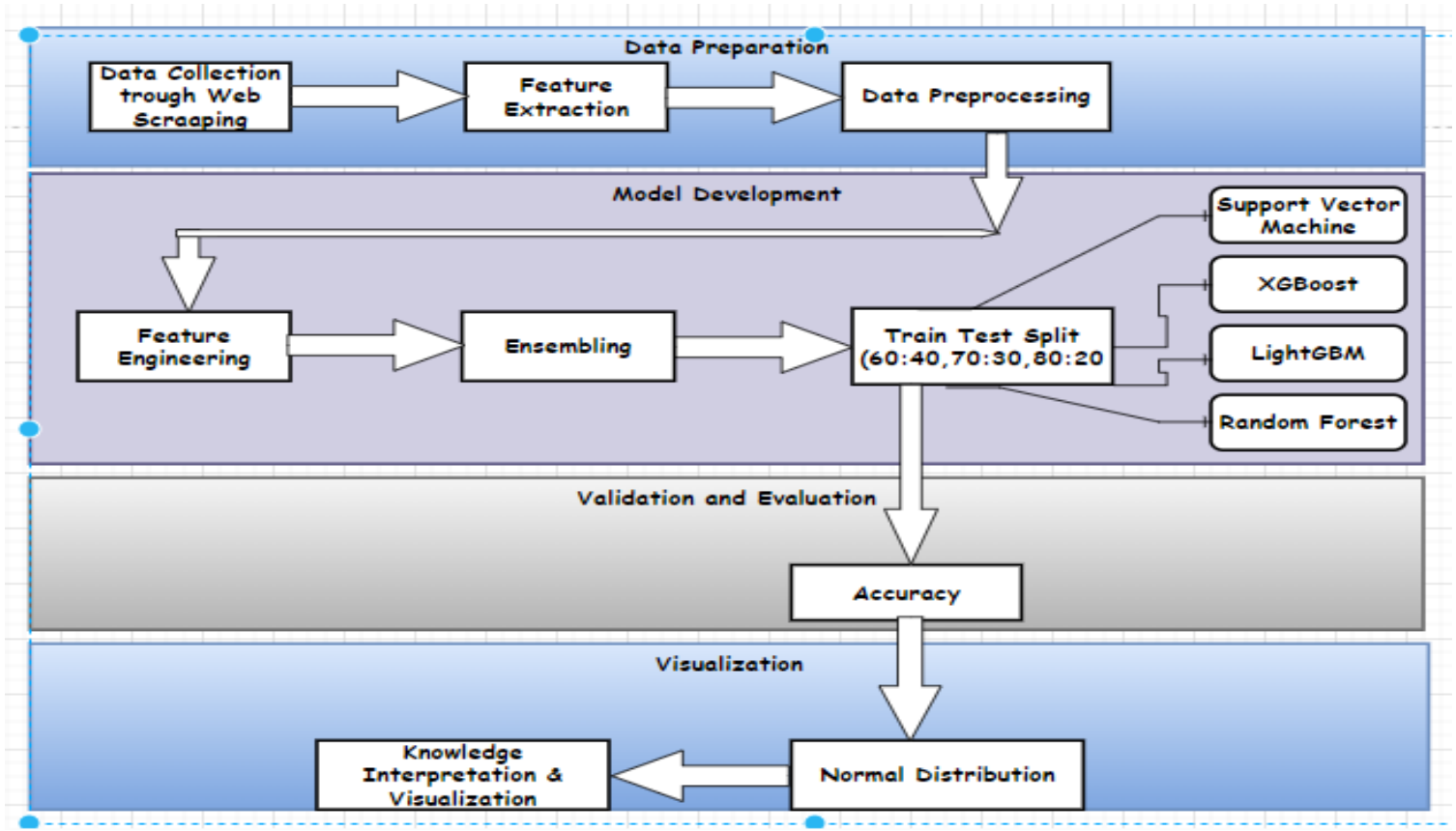


1. **To carry out review of domain and technology required for big data analytics**
2. **To capture the data required for generic drug selection from various resources**
3. **To design intelligent predictive model for right generic drug selection in pharmaceutical industry.**
4. **To implement the designed model using big data analytics**

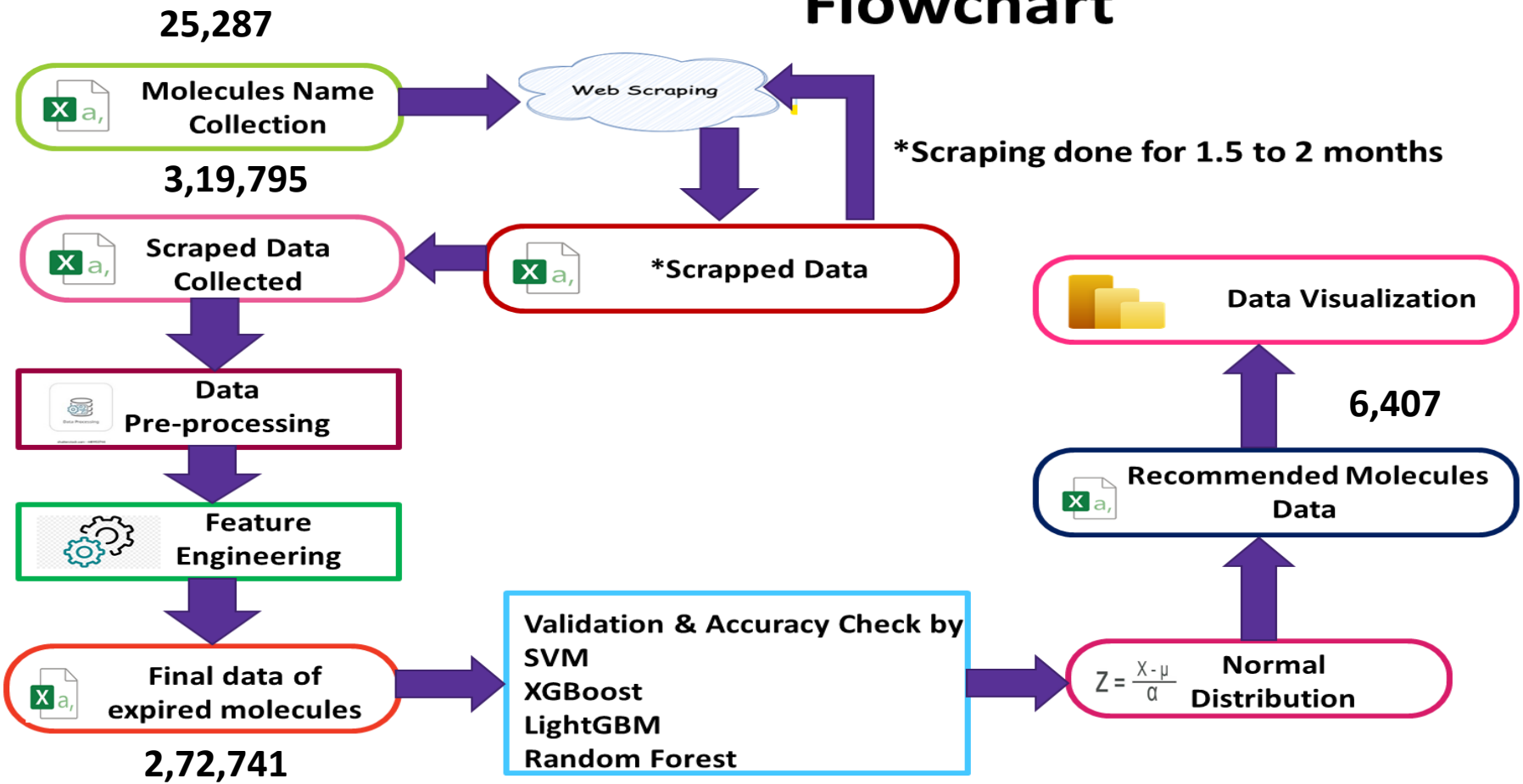
Design


Define Objectives
Data Collection
Data Preprocessing
EDA
Model Selection
Train Model
Test Model
Implement with Dashboard

Model Diagram



Flowchart


25,287
Molecules Name Collection
Web Scraping
*Scraping done for 1.5 to 2 months
3,19,795
Scraped Data Collected
*Scrapped Data
Data Visualization
Data Pre-processing
6,407
Recommended Molecules Data
Feature Engineering
Validation & Accuracy Check by
SVM
XGBoost
LightGBM
Random Forest
Z = X - μ / α
Normal Distribution
Final data of expired molecules
2,72,741

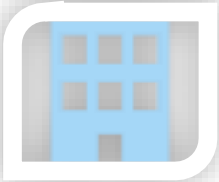
Data Collection

To take a decision of right generic drug selection data collected online by considering following parameters.

Applicant Holder



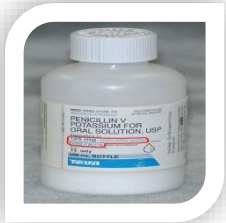
Sector



Dosage



Strength



Approval Date



Route



Active Ingredient



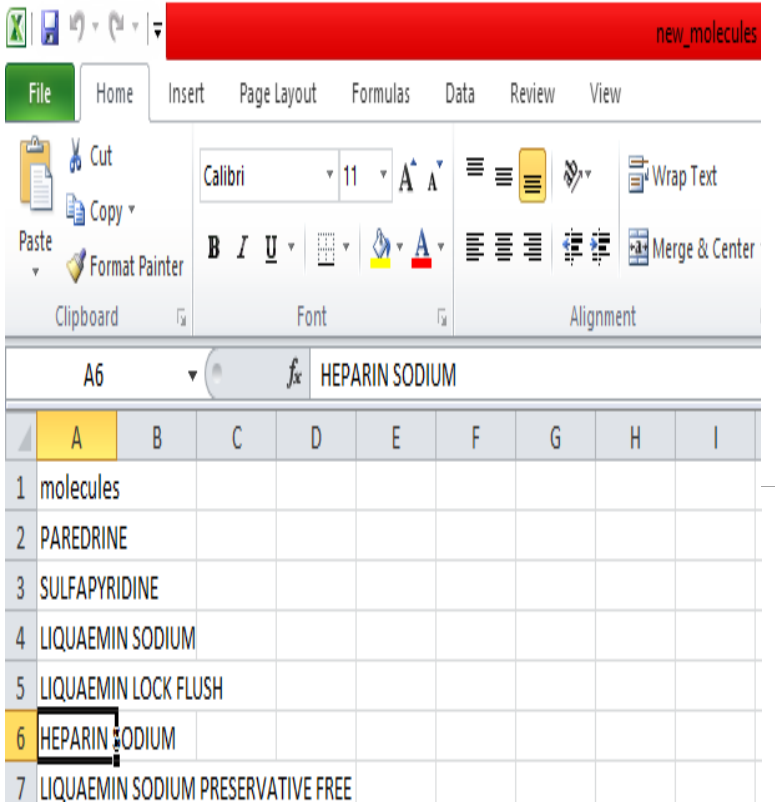
Most of the data available online through orange book

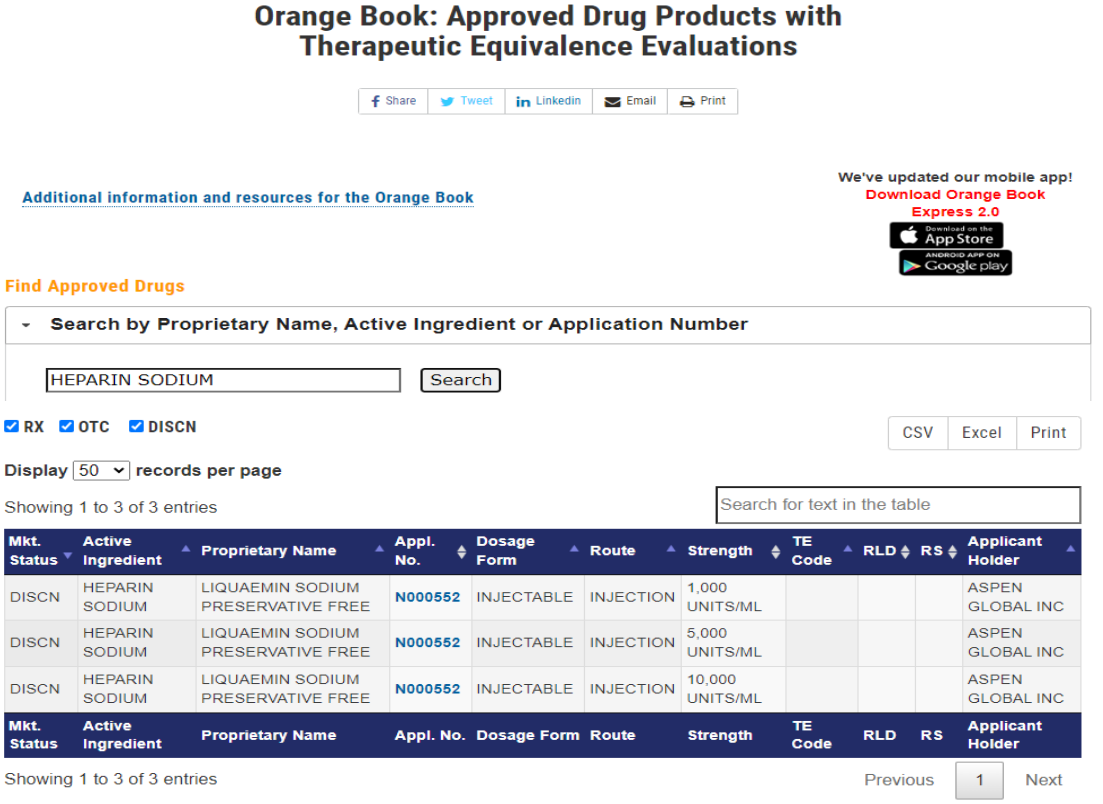
Orange Book: Approved Drug Products with Therapeutic Equivalence Evaluations (fda.gov)

- https://www.drugs.com/dosage/azithromycin.html
- Notification and Report on National List of Essential Medicines, 2022.pdf (mohfw.gov.in)
- https://www.pdfhunter.com/wp-content/uploads/2021/08/Jan-Aushadhi-Medicine-List-2021-PDF.pdf
- https://purplebooksearch.fda.gov/downloads
- https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=browseByLetter.page&productLetter=A&ai=0

Input->Process->Output

Input





Mkt. Status	Active Ingredient	Proprietary Name	Appl. No.	Dosage Form	Route	Strength	TE Code	RLD	RS	Applicant Holder
DISCN	HEPARIN SODIUM	LIQUAEMIN SODIUM PRESERVATIVE FREE	N000552	INJECTABLE	INJECTION	1,000 UNITS/ML				ASPEN GLOBAL INC
DISCN	HEPARIN SODIUM	LIQUAEMIN SODIUM PRESERVATIVE FREE	N000552	INJECTABLE	INJECTION	5,000 UNITS/ML				ASPEN GLOBAL INC
DISCN	HEPARIN SODIUM	LIQUAEMIN SODIUM PRESERVATIVE FREE	N000552	INJECTABLE	INJECTION	10,000 UNITS/ML				ASPEN GLOBAL INC
Mkt. Status	Active Ingredient	Proprietary Name	Appl. No.	Dosage Form	Route	Strength	TE Code	RLD	RS	Applicant Holder

Continue…

combined_results.csv

Active Ingredient	Dosage Form	Route	Strength	Applicant Holder	Approval Date	Search Term
HEPARIN SODIUM	INJECTABLE	INJECTION	200 UNITS/100ML	MCGAW INC	Dec 31, 1984	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	2,000 UNITS/100ML	BAXTER HEALTHCARE CORP	Jul 9, 1985	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	10,000 UNITS/100ML	HOSPIRA INC	Jan 30, 1985	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	5,000 UNITS/100ML	HOSPIRA INC	Jan 30, 1985	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	200 UNITS/100ML	MCGAW INC	Dec 31, 1984	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	4,000 UNITS/100ML	BAXTER HEALTHCARE CORP	Oct 31, 1983	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	4,000 UNITS/100ML	HOSPIRA INC	Jan 25, 1989	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	5,000 UNITS/100ML	BAXTER HEALTHCARE CORP	Jul 9, 1985	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	10,000 UNITS/100ML	BAXTER HEALTHCARE CORP	Jul 2, 1987	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	5,000 UNITS/100ML	HOSPIRA INC	Jan 30, 1985	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	10,000 UNITS/100ML	HOSPIRA INC	Jan 30, 1985	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	5,000 UNITS/100ML	B BRAUN MEDICAL INC	Mar 29, 1985	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	5,000 UNITS/100ML	HOSPIRA INC	Jan 25, 1989	DEXTROSE
HEPARIN SODIUM	INJECTABLE	INJECTION	1,000 UNITS/100ML	MCGAW INC	Dec 31, 1984	DEXTROSE
LEVOFLOXACIN	INJECTABLE	INJECTION	EQ 250MG/50ML (EQ 5MG/ML)	JANSSEN PHARMACEUTICALS INC	Dec 20, 1996	DEXTROSE
LEVOFLOXACIN	INJECTABLE	INJECTION	EQ 500MG/100ML (EQ 5MG/ML)	JANSSEN PHARMACEUTICALS INC	Dec 20, 1996	DEXTROSE
LEVOFLOXACIN	INJECTABLE	INJECTION	EQ 750MG/150ML (EQ 5MG/ML)	JANSSEN PHARMACEUTICALS INC	Dec 20, 1996	DEXTROSE
LEVOFLOXACIN	INJECTABLE	INJECTION	EQ 250MG/50ML (EQ 5MG/ML)	EUGIA PHARMA SPECIALITIES LTD	Feb 10, 2016	DEXTROSE
LEVOFLOXACIN	INJECTABLE	INJECTION	EQ 500MG/100ML (EQ 5MG/ML)	EUGIA PHARMA SPECIALITIES LTD	Feb 10, 2016	DEXTROSE
LEVOFLOXACIN	INJECTABLE	INJECTION	EQ 750MG/150ML (EQ 5MG/ML)	EUGIA PHARMA SPECIALITIES LTD	Feb 10, 2016	DEXTROSE
LIDOCAINE HYDROCHLORIDE	INJECTABLE	INJECTION	100MG/100ML	BAXTER HEALTHCARE CORP	Approved Prior to Jan 1, 1982	DEXTROSE
LIDOCAINE HYDROCHLORIDE	INJECTABLE	INJECTION	200MG/100ML	B BRAUN MEDICAL INC	Mar 30, 1984	DEXTROSE
LIDOCAINE HYDROCHLORIDE	INJECTABLE	INJECTION	200MG/100ML	HOSPIRA INC	Approved Prior to Jan 1, 1982	DEXTROSE

Process

Data Preprocessing- Removal of Noisy, Duplicate, Missing ,Null Data and dropping of columns

final_data.csv

Out[3]:

	Active Ingredient	Dosage Form	Route	Strength	Applicant Holder	Approval Date	Search Term
0	AMILORIDE HYDROCHLORIDE; HYDROCHLOROTHIAZIDE	TABLET	ORAL	EQ 5MG ANHYDROUS; 50MG	BARR LABORATORIES INC	May 10, 1988	PAREDRINE
1	AMILORIDE HYDROCHLORIDE; HYDROCHLOROTHIAZIDE	TABLET	ORAL	EQ 5MG ANHYDROUS; 50MG	RISING PHARMA HOLDING INC	Oct 31, 1991	PAREDRINE
2	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	10MG; 2MG	MYLAN PHARMACEUTICALS INC	Nov 10, 1988	PAREDRINE
3	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	10MG; 4MG	MYLAN PHARMACEUTICALS INC	Nov 10, 1988	PAREDRINE
4	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	25MG; 2MG	MYLAN PHARMACEUTICALS INC	Nov 10, 1988	PAREDRINE

Activate Windows

Out[5]:

	Active Ingredient	Dosage Form	Route	Strength	Applicant Holder	Approval Date
0	AMILORIDE HYDROCHLORIDE; HYDROCHLOROTHIAZIDE	TABLET	ORAL	EQ 5MG ANHYDROUS; 50MG	BARR LABORATORIES INC	May 10, 1988
1	AMILORIDE HYDROCHLORIDE; HYDROCHLOROTHIAZIDE	TABLET	ORAL	EQ 5MG ANHYDROUS; 50MG	RISING PHARMA HOLDING INC	Oct 31, 1991
2	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	10MG; 2MG	MYLAN PHARMACEUTICALS INC	Nov 10, 1988
3	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	10MG; 4MG	MYLAN PHARMACEUTICALS INC	Nov 10, 1988
4	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	25MG; 2MG	MYLAN PHARMACEUTICALS INC	Nov 10, 1988

Continue…

Feature Engineering-Addition of new column Sector ,Performing manipulation on column

.2]:

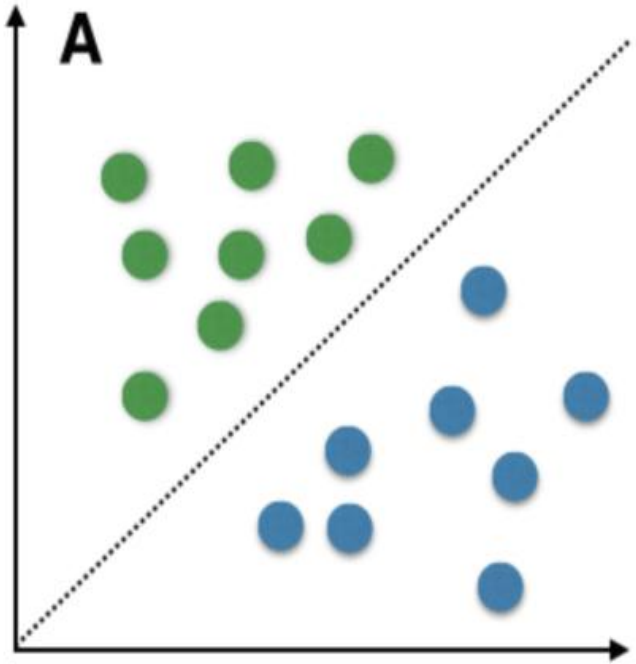
	Active Ingredient	Dosage Form	Route	Strength	Applicant Holder	Approval Date	Sector	Status
0	AMILORIDE HYDROCHLORIDE; HYDROCHLOROTHIAZIDE	TABLET	ORAL	EQ 5MG ANHYDROUS; 50MG	BARR LABORATORIES INC	1988-05-10	RETAIL	Expired
1	AMILORIDE HYDROCHLORIDE; HYDROCHLOROTHIAZIDE	TABLET	ORAL	EQ 5MG ANHYDROUS; 50MG	RISING PHARMA HOLDING INC	1991-10-31	RETAIL	Expired
2	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	10MG; 2MG	MYLAN PHARMACEUTICALS INC	1988-11-10	RETAIL	Expired
3	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	10MG; 4MG	MYLAN PHARMACEUTICALS INC	1988-11-10	RETAIL	Expired
4	AMITRIPTYLINE HYDROCHLORIDE; PERPHENAZINE	TABLET	ORAL	25MG; 2MG	MYLAN PHARMACEUTICALS INC	1988-11-10	RETAIL	Expired

Continue…

Support Vector Machine (SVM)

- Ideal For- Binary classification
- Definition-Supervised machine learning algorithm for **classification**
- The main idea behind SVM is to find the optimal boundary (called a **hyper plane**) that best separates the data points of different classes.
- Output:

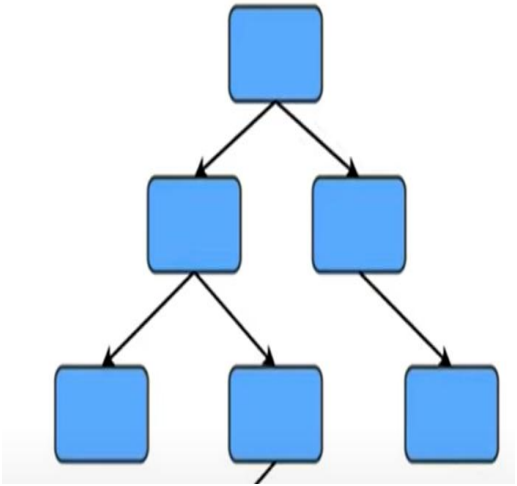
	Train Size	SVM Accuracy
0	0.6	78.29
1	0.7	78.37
2	0.8	78.68



Continue…
eXtreme Gradient Boosting(XGBoost)

- Ideal For- Classification, large dataset
- Definition-A powerful and efficient open-source gradient boost ing algorithm designed for speed and performance. Widely use d for structured/tabular data.
- The main idea behind XGBoost is it builds an ensemble of trees that passed features to trained trees each tree returns prediction. Add up of all tress to generate final score.
- Output:

	Train Size	XGBoost Accuracy
0	0.6	89.54
1	0.7	89.46
2	0.8	89.61

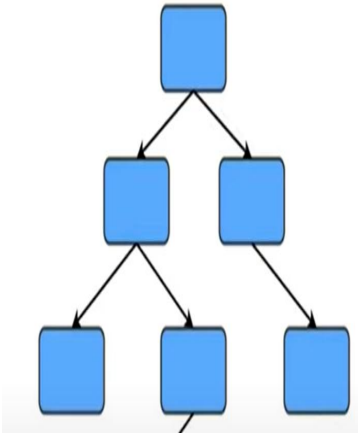


Continue…

Light GBM (Gradient Boosting Method)

- Ideal For- Large datasets, high performance, low memory usage high accuracy
- Definition: Fast, distributed, and efficient gradient boosting framework that uses tree-based learning algorithms to solve classification problem on large datasets.
- The main idea behind using Light GBM in my model because it can directly handles categorical data without one hot encoding.
- Output:

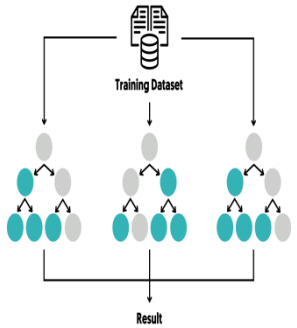
	Train Size	LightGBM Accuracy
0	0.6	84.97
1	0.7	84.92
2	0.8	85.26



Continue…

Random Forest

- Ideal For- Handling many features , reduces over fitting
- Definition: It is an ensemble learning method that combines the predictions of many decision trees, each built on a random subset of the data and features, to improve accuracy and reduce over fitting.
- The main idea behind using this algorithm in my research because it is robust and given the highest accuracy with the help of voting classifier.
- Output:



	Train Size	RandomForest Accuracy
0	0.6	95.34
1	0.7	95.90
2	0.8	96.16

Output

Label Encoding, Validations and finding accuracies through ML Models

Out[65]:

	Train Size	SVM Accuracy	XGBoost Accuracy	LightGBM Accuracy	RandomForest Accuracy
0	0.6	78.29	89.54	84.97	95.34
1	0.7	78.37	89.46	84.92	95.90
2	0.8	78.68	89.61	85.26	96.16

Continue…

Z=X-μ/σ
X-Data Point
μ- Average Applicant Holder count
σ- Standard Deviation

If z-score>1.0
Player count:167 to 123
Grade: Highly Not Recommended
If z-score> 0.5
Player count:122 to 99
Grade: Not Recommended
If z-score>-0.5
Player count:98 to 52
Grade: Average
If z-score>-1.0
Player count:51 to 28
Grade: Can Recommend
If z-score<-1.0
Player count:27 to 01
Grade: Highly Recommended

```
def assign_grade(z):
    if z > 1.0:
        return 'Highly Not Recommended'
    elif z > 0.5:
        return 'Not Recommended'
    elif z > -0.5:
        return 'Average'
    elif z > -1.0:
        return 'Can Recommend'
    else:
        return 'Highly Recommended'

Expired_molecules['grade'] = Expired_molecules['z_score'].apply(assign_grade)

# Step 5: Save the DataFrame with the new grade column back to a CSV file
Expired_molecules.to_csv('expired_products.csv', index=False)
```

```
min_count_holder = holder_counts.idxmin()
max_count_holder = holder_counts.idxmax()

print(f"Total row count: {total_count}")
print(f"Total distinct count: {total_distinct_count}")
print(f"Average count: {average_count}")
print(f"Min count: {min_count} Holder: {min_count_holder}")
print(f"Max count: {max_count} Holder: {max_count_holder}")
std_dev = np.std(holder_counts)
print(std_dev)
```

```
Total row count: 272741
Total distinct count: 1774
Average count: 153.7435174746336
Min count: 1 Holder: UNIV ALABAMA AT BIRMINGHAM
Max count: 6608 Holder: WATSON LABORATORIES INC
507.4745528746843
```

```
below_average_holders = holder_counts[holder_counts < average_count].index
filtered_df = data[data['Applicant Holder'].isin(below_average_holders)]
```

Continue…

Apply Normal Distribution as a key parameter in Decision Making

	Active Ingredient	Dosage Form	Route	Strength	Applicant Holder	Approval Date	Sector	Status	Applicant Holder count	z_score	grade
19785	DEXBROMPHENIRAMINE MALEATE; PSEUDOEPHEDRINE SU...	TABLET, EXTENDED RELEASE	ORAL	6MG; 120MG **Federal Register determination th...	SCHERING PLOUGH HEALTHCARE PRODUCTS INC	1982-09-13	RETAIL	Expired	44	-0.575059	Can Recommend
19786	DEXBROMPHENIRAMINE MALEATE; PSEUDOEPHEDRINE SU...	TABLET, EXTENDED RELEASE	ORAL	6MG; 120MG	PIONEER PHARMACEUTICALS INC	1988-06-16	RETAIL	Expired	131	1.442931	Highly Not Recommended
19787	EPHEDRINE SULFATE	SOLUTION	INTRAVENOUS	50MG/ML (50MG/ML)	RENEW PHARMACEUTICALS LTD	2017-03-01	HOSPITAL	Expired	11	-1.340504	Highly Recommended
19791	DELAFLOXACIN MEGLUMINE	POWDER	INTRAVENOUS	EQ 300MG BASE/VIAL	MELINTA SUBSIDIARY CORP	2017-06-19	HOSPITAL	Expired	8	-1.410090	Highly Recommended
19792	DELAFLOXACIN MEGLUMINE	TABLET	ORAL	EQ 450MG BASE	MELINTA SUBSIDIARY CORP	2017-06-19	RETAIL	Expired	8	-1.410090	Highly Recommended

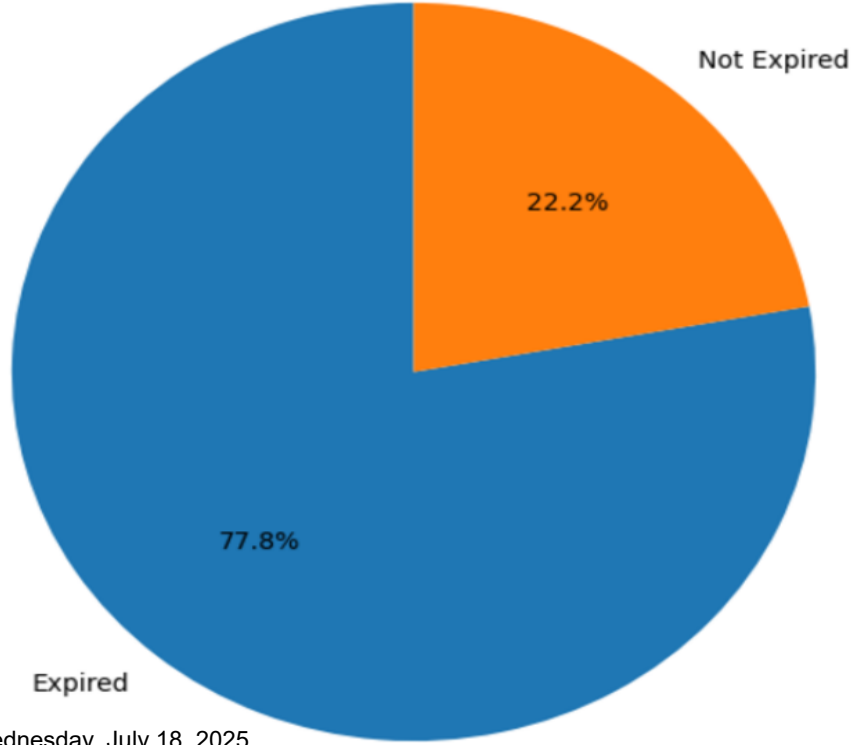
Continue…

Expired_molecule.csv

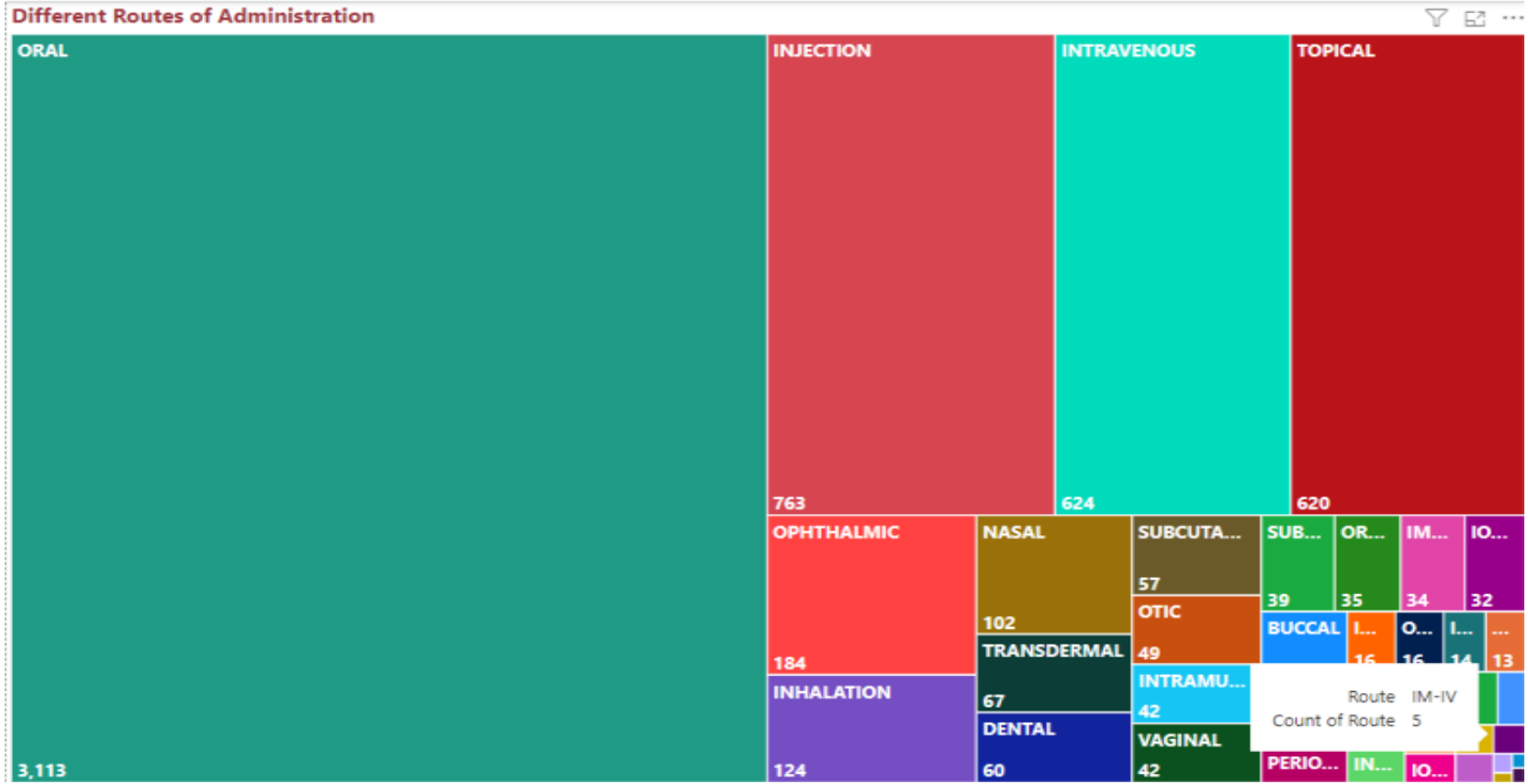
	Active Ingredient	Dosage Form	Route	Strength	Applicant Holder	Approval Date	Sector	Status	Applicant Holder count	z_score	grade
155	CABAZITAXEL	SOLUTION	INTRAVENOUS	60MG/1.5ML (40MG/ML)	SANOFI AVENTIS US INC	2010-06-17	HOSPITAL	Expired	7	-1.433285	Highly Recommended
303	FLUPHENAZINE DECANOATE	INJECTABLE	INJECTION	25MG/ML	PAR STERILE PRODUCTS LLC	2014-07-03	HOSPITAL	Expired	21	-1.108551	Highly Recommended
367	HYDRALAZINE HYDROCHLORIDE	INJECTABLE	INJECTION	20MG/ML	BEOWULF ASSET MANAGEMENT LP	2015-06-29	HOSPITAL	Expired	1	-1.572457	Highly Recommended
671	OLSALAZINE SODIUM	CAPSULE	ORAL	250MG	MYLAN SPECIALTY LP A VIATRIS CO	1990-07-31	RETAIL	Expired	14	-1.270918	Highly Recommended

Data Visualization-Pre Visualization

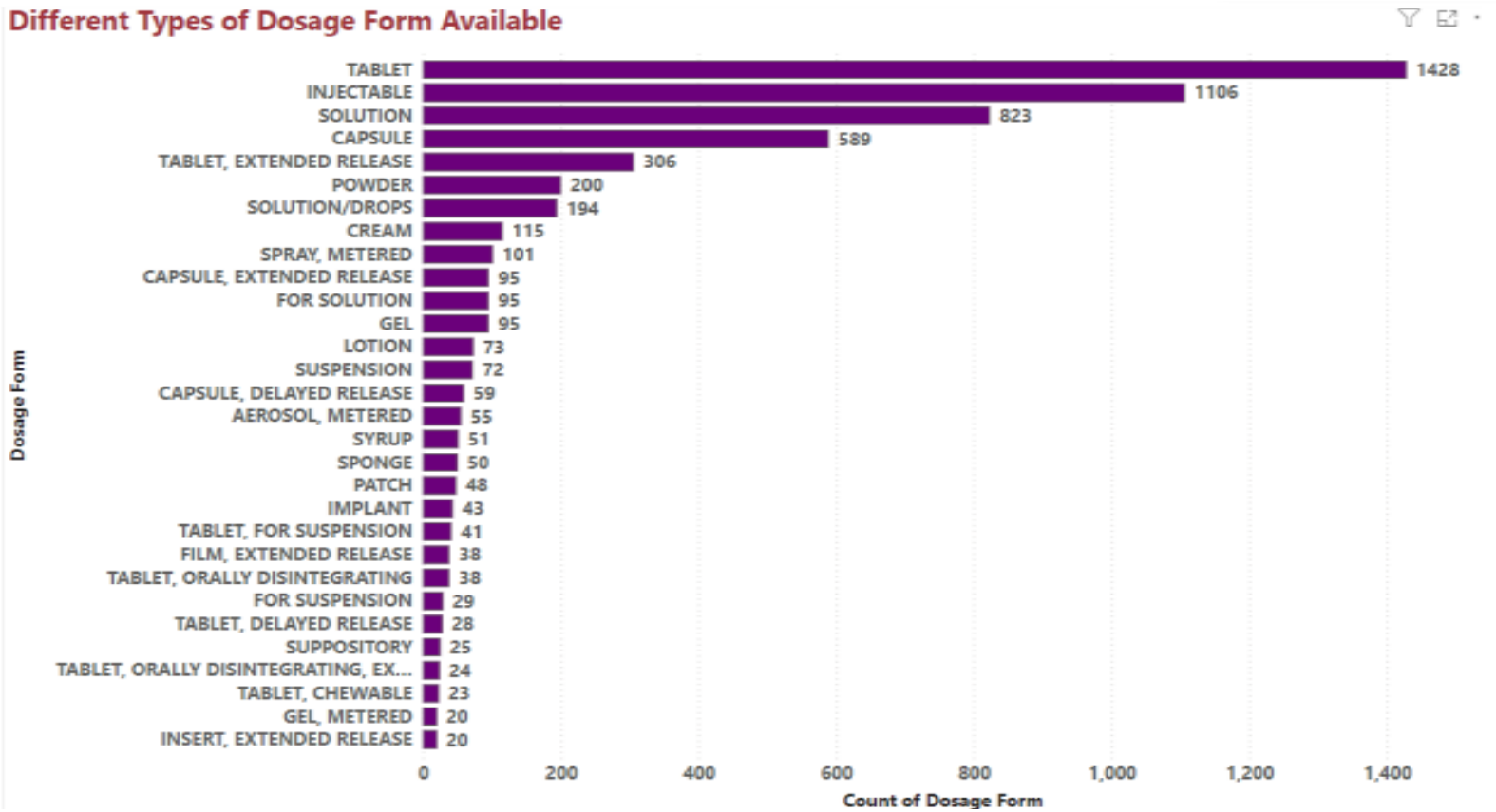
Expired and Not expired molecules



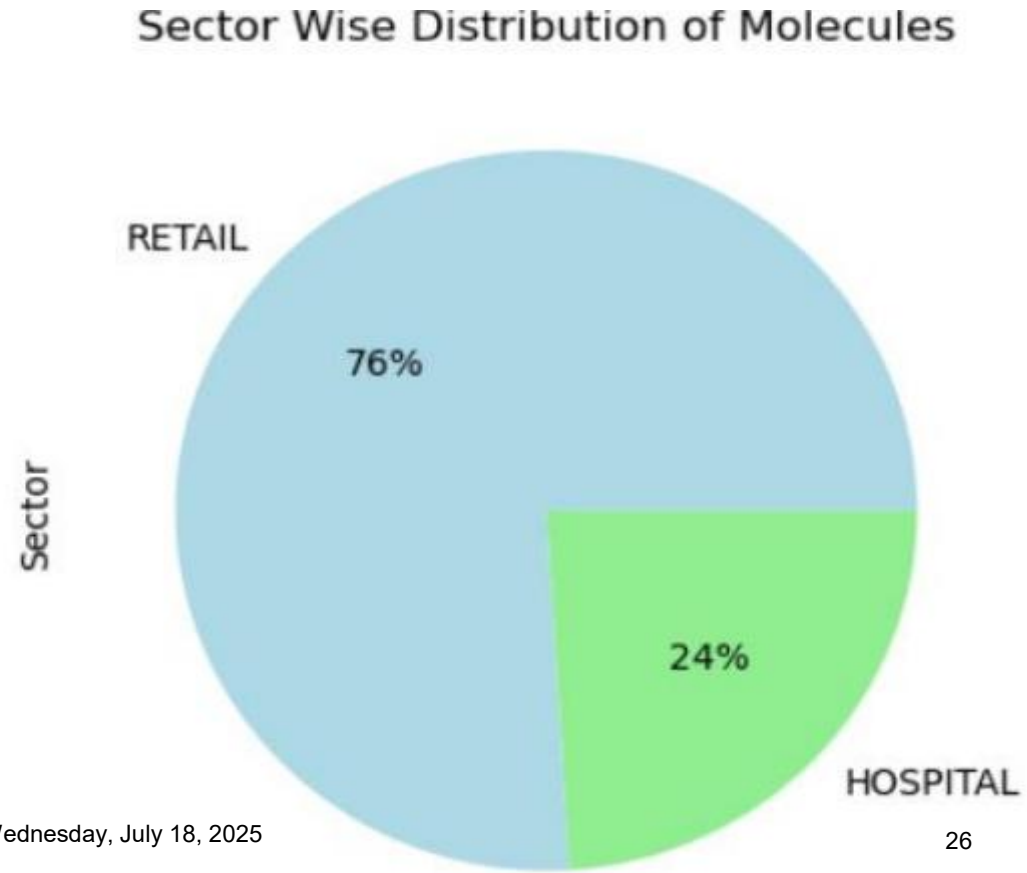
Different Routes of Administration of a molecule



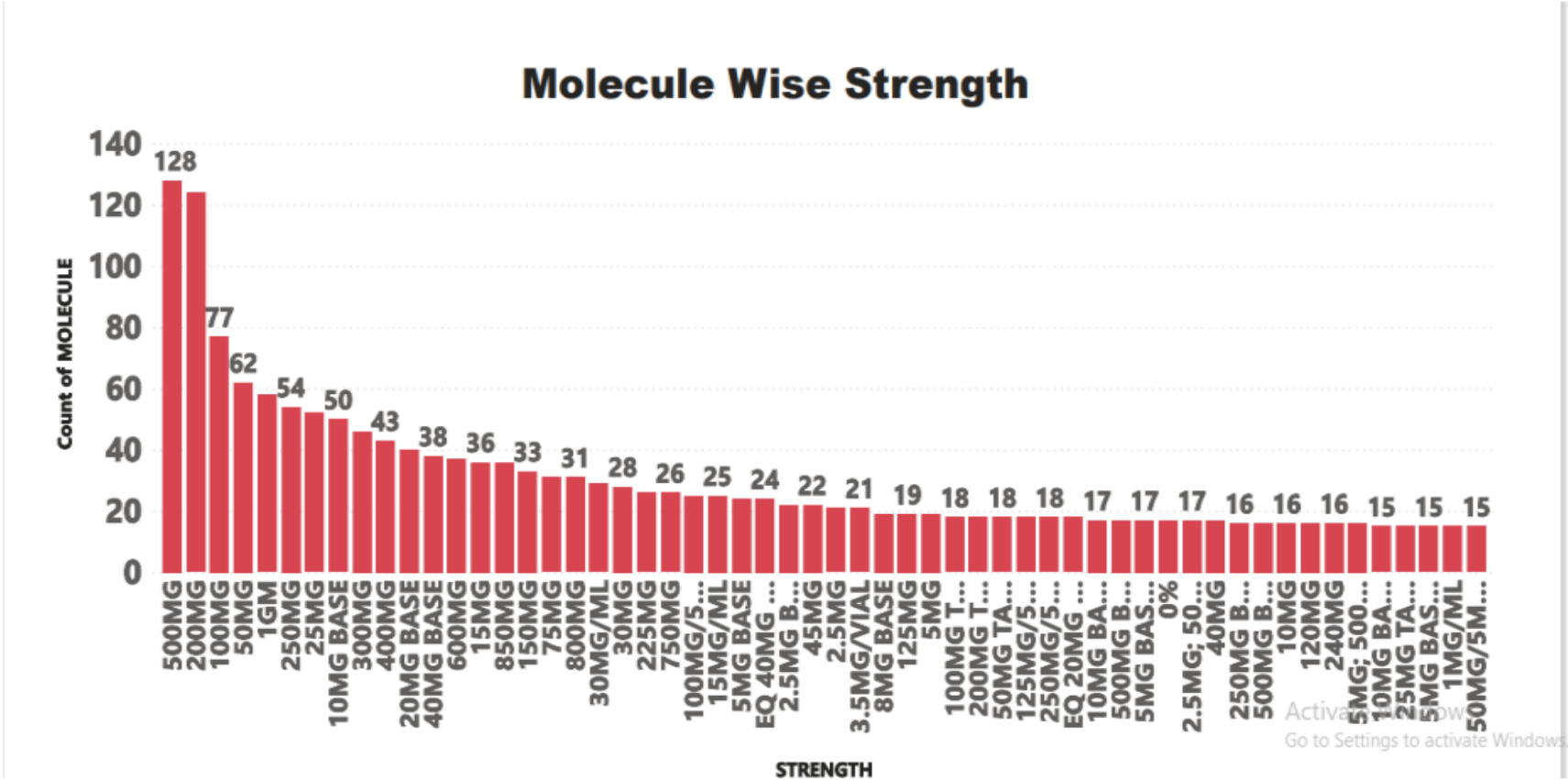
Different Dosage forms available



Sector Wise Distribution of Molecules

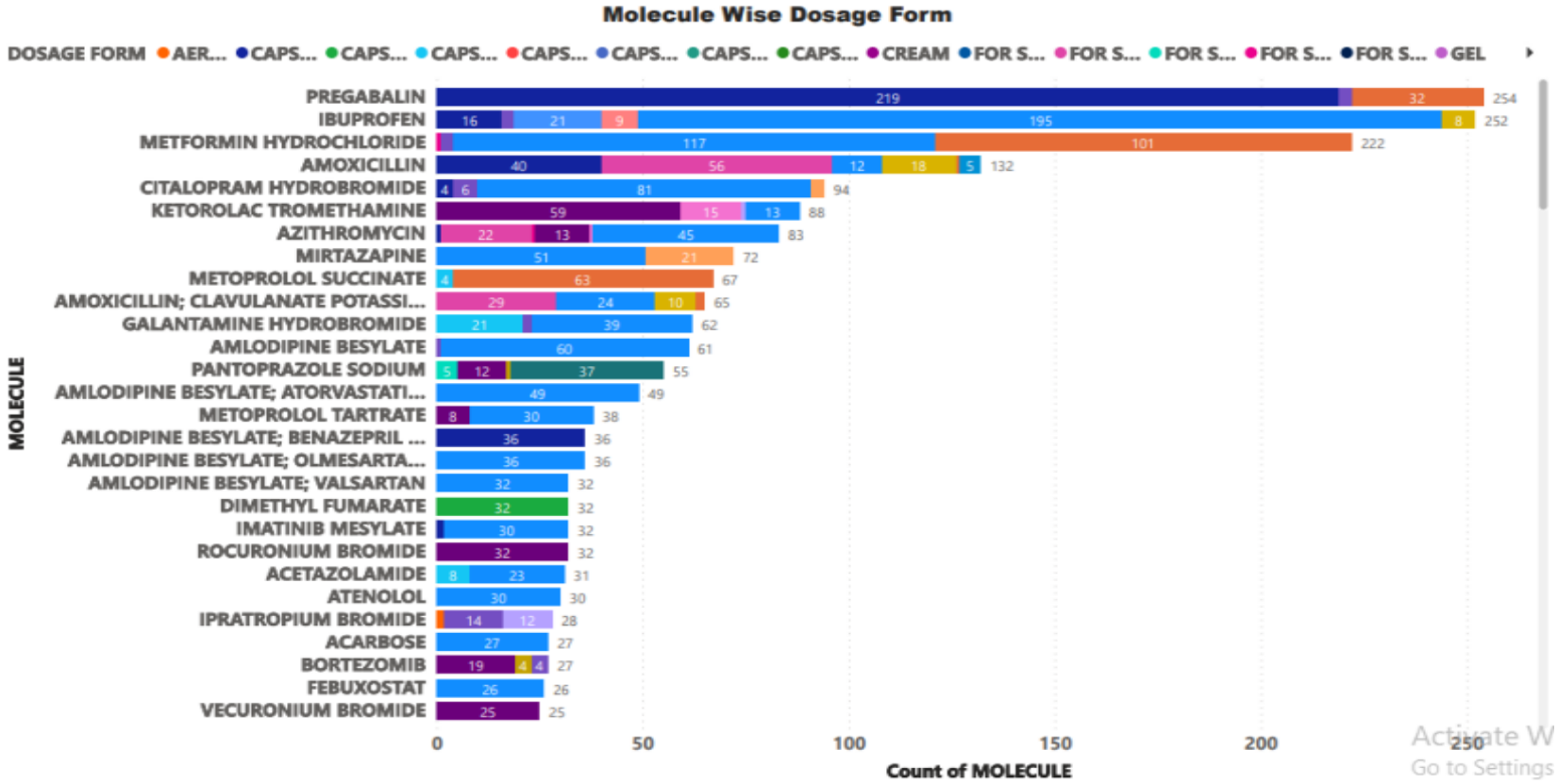


Molecule Wise strengths available

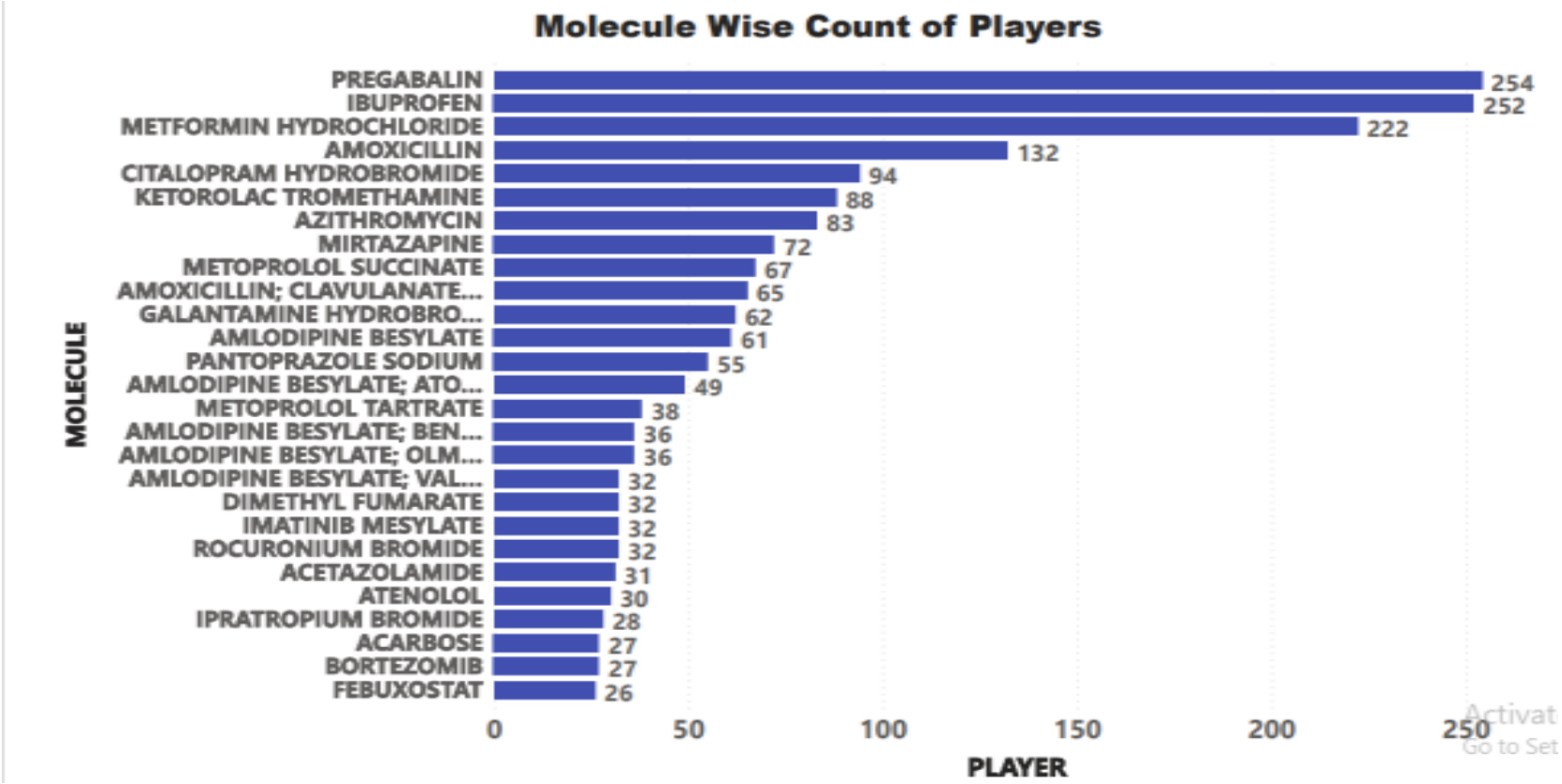


Activate Windows
Go to Settings to activate Windows.

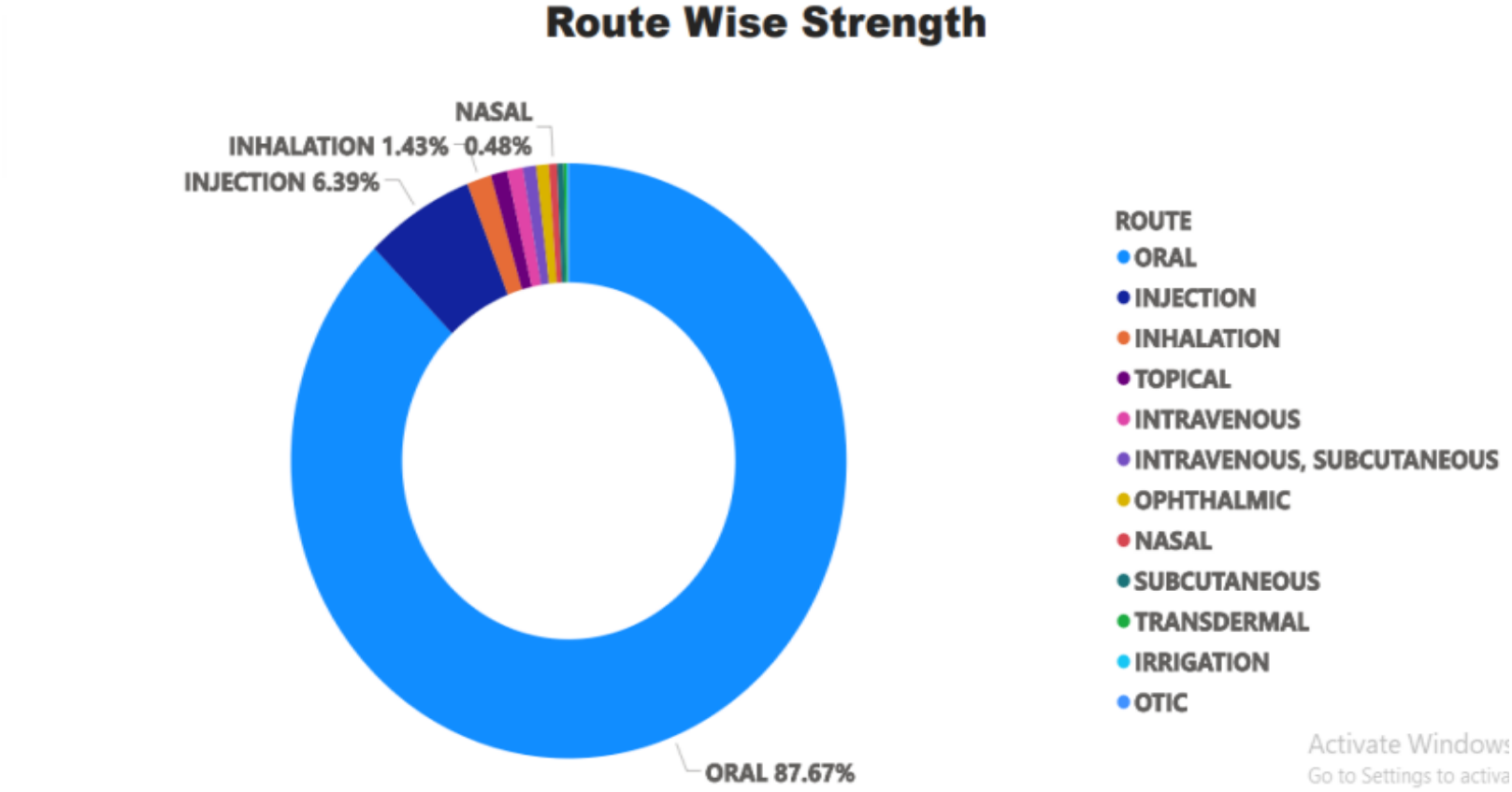
Molecule Wise Dosage Form



Molecule Wise Count of Players

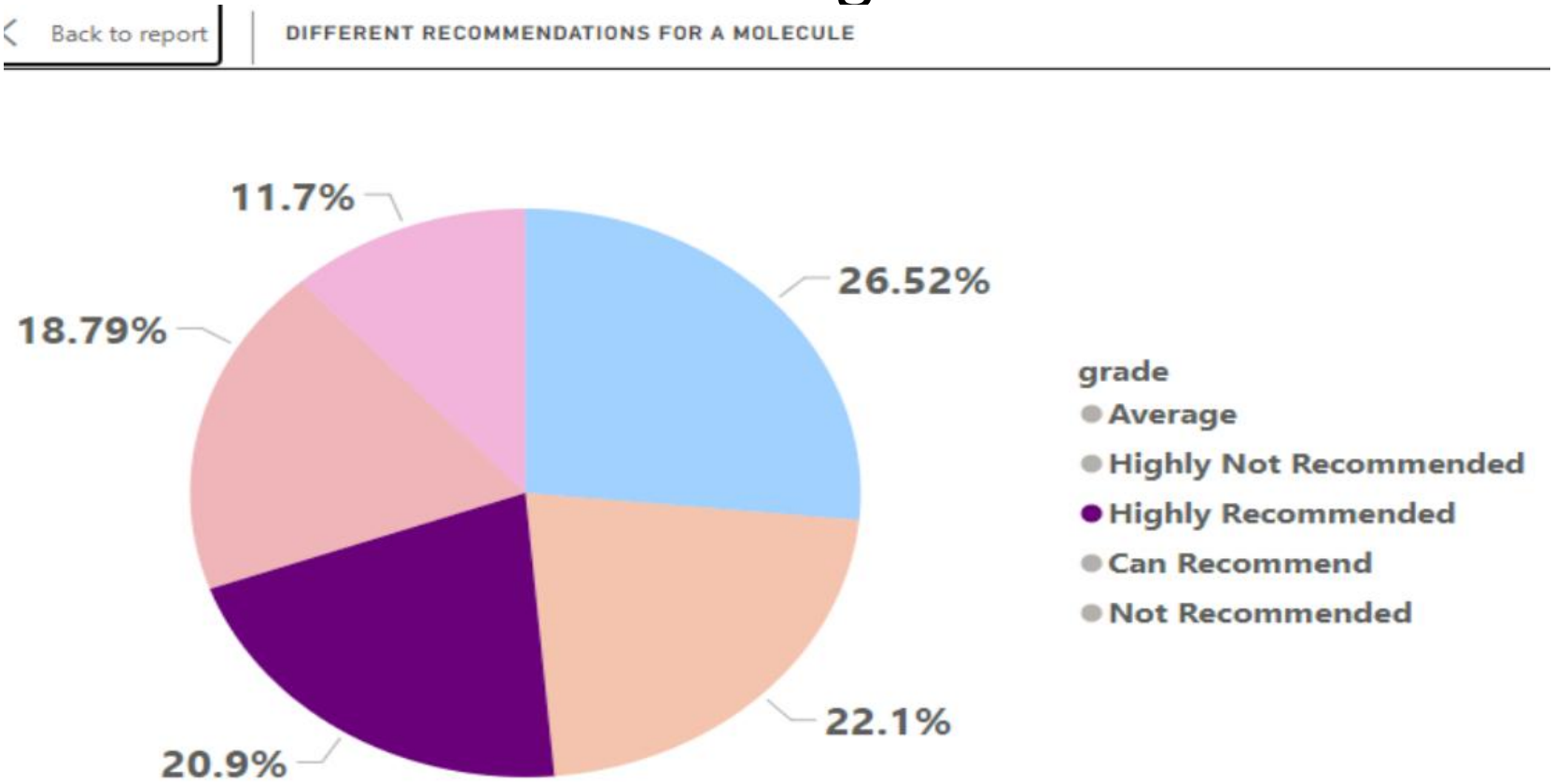


Route Wise Strength



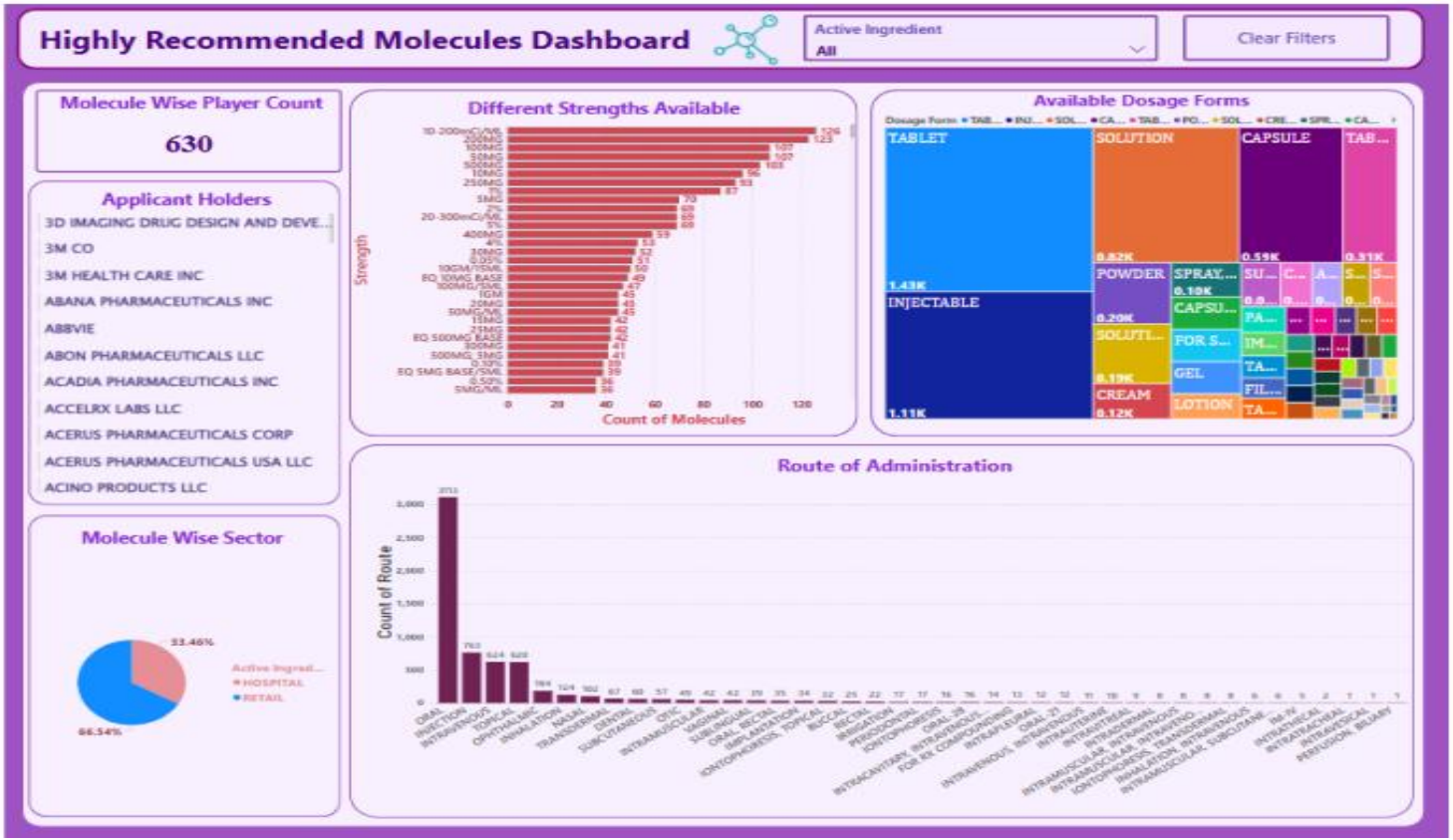
Activate Windows
Go to Settings to activa

Different Recommendations generated for a molecule



Post Visualization

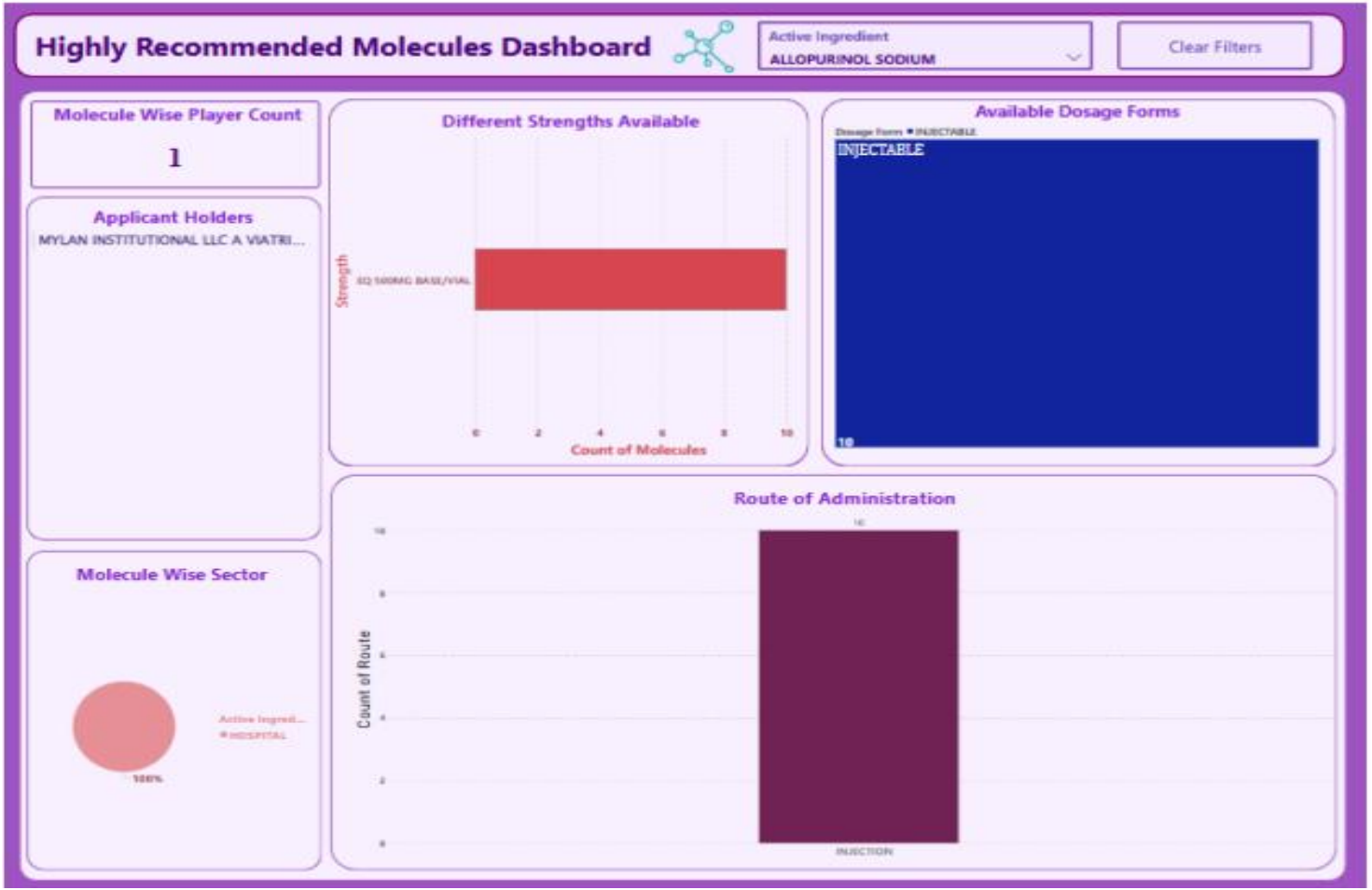
Highly Recommended Molecules Dashboard



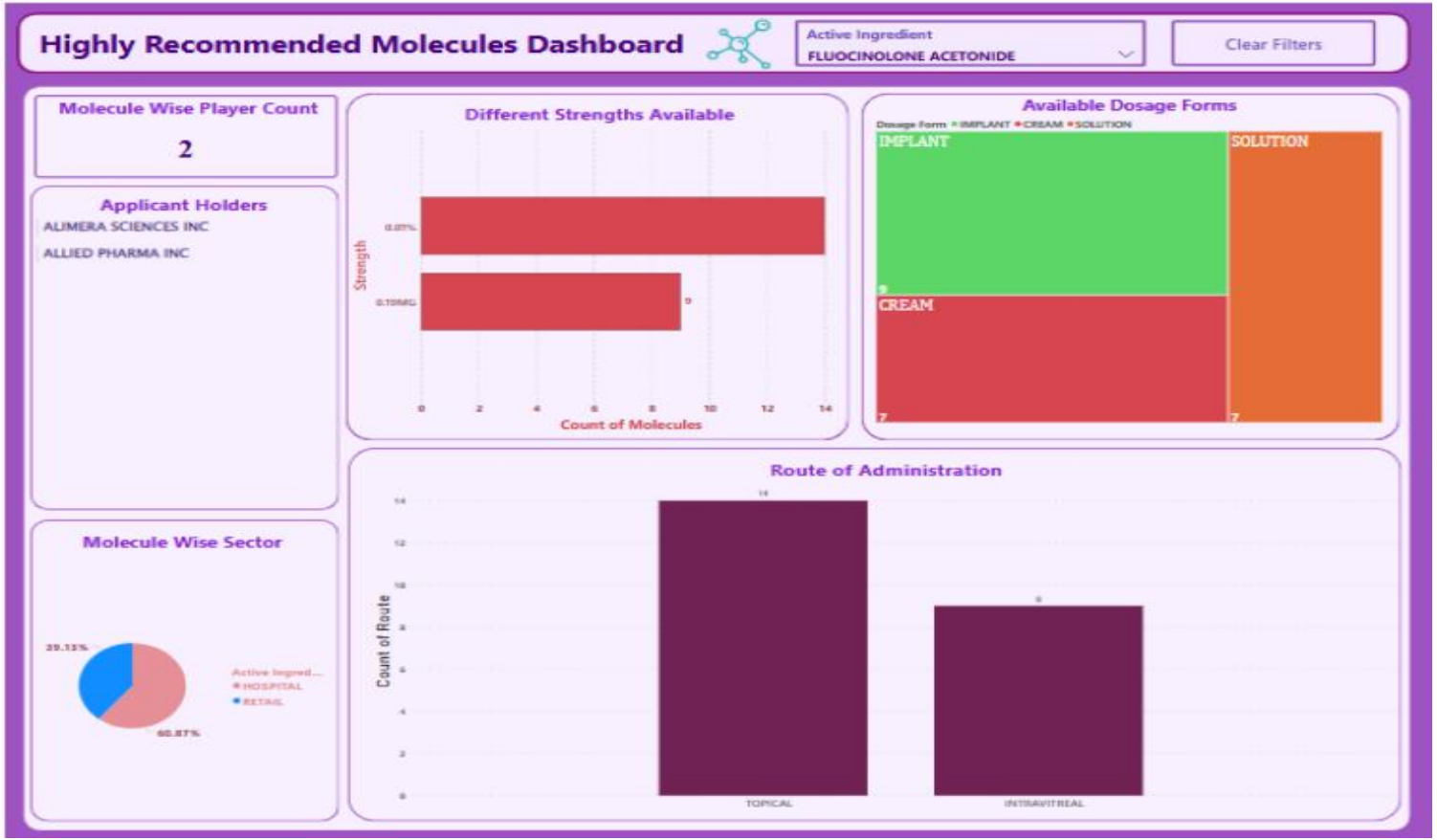
Molecule with less players and applicable sector is retail



A Molecule with less Player and Applicable Sector is Hospital Only



A molecule with only two players and applicable sector is Hospital and Retail



Conclusion

- The model takes input as molecule names and extracts relevant features using the Selenium web scraping technique
- The expired molecules were taken into consideration and are stored.
- Exploratory Data Analysis (EDA) was carried out to clean and preprocess data.
- Feature engineering was performed to manipulate features, such as the approval date, and to create new features based on existing ones e.g. a new feature called “Sector” was derived from the “Route” feature
- Using various Python functions, the model was capable of determining the Applicant Holder-wise count of molecules to establish a threshold value for each Applicant Holder.
- Multiple machine learning algorithms, including Support Vector Machine (SVM),XGBoost, LightGBM, and Random Forest, were employed to enhance the performance and robustness of the developed model. The dataset was trained and tested using different training and testing splits (60:40, 70:30, and 80:20) to improve the models performance, accuracy, and reliability. The model demonstrated optimal performance with the 80:20 split, achieving an accuracy of 96.16%
- The data was validated using the K-fold cross-validation technique.
- Through statistical method of a normal distribution z-score values were generated to classify a molecule into categories such as Highly Recommended, Can Recommend, Average Recommended, Not Recommended, Highly Not Recommended.
- The model extracts and presents all Highly Recommended molecules, serving as a critical tool for decision-makers in the selection process of the appropriate molecule.
- The model utilizes Power BI as an effective tool for representing pre visualization as well as post visualizations of the Highly Recommended molecules data.

Publications

- Intelligent Predictive Modeling Using Big Data for Drug Selection in Pharmaceutical Industry (Published & Presented)

Details of the first paper	Description of the first paper
• Presented and Published above paper • Scopus Publication Link- • https://www.turcomat.org/index.php/turkbilmat/article/view/8087/6327 • Presented this paper in International conference IC2TMA at IMCC	Innovative and Generic drugs DP and API Drug selection methodology 4V's of Big Data Big Data fro drug selection Strength of predictive analytics Big Data Analytical Tools like Hadoop,Python,R, etc.

Continue…

- Role of Big Data Analysis in Pharmaceutical Product Selection (Published)

Details of the second paper	Description of the second paper
• https://drive.google.com/file/d/12EgwmydElVHAReSiV8cL4eCcGRl4z8mE/view • Published this paper in Gradiva Review Journal	Big Data Analytics Types of Big Data Analytics Data Analysis Steps

Continue…

- Study of big data analytical open source tools: A roadmap for pharmaceutical generic drug selection (Published & Presented)

Details of the third paper	Description of the third paper
• Presented and published this paper online in International Conference on Recent Trends and Advancement in Science, Technology, Commerce and Management at Sanjivani College of Engineering, Kopergoan in association with Jorden River University USA . • UGC publication Link • file:///E:/Netraja%20Data/D/Ph.%20D.%20data%20NCM/six%20monthly%20report%202/third%20research%20paper%20e-book.pdf	Introduction Importance of Data Analytics in Pharmaceutical Companies for right selection of product Big Data Analytical Tools

Continue…

❑ Why Big Data for Pharmaceutical Product Selection: A Review

Details of the fourth paper	Description of the fourth paper
• Published this paper in UGC Care Journal named Fuzzy Systems and Soft Computing . • Paper Publication Link: • PAPER(19) (1).pdf	As large volume of pharmaceutical generic drug data cannot be managed by traditional databases. So this paper focuses on how 4 V ’s of big data i.e. Volume, Variety, Velocity Veracity will fit into parameters of generic drug selection.

Summary of Research Papers

Sr. No.	Name	Year
1.	Intelligent Predictive Modeling Using Big Data for Drug Selection in Pharmaceutical Industry	2021
2.	Role of Big Data Analysis in Pharmaceutical Product Selection	2022
3.	Study of big data analytical open source tools: A roadmap for pharmaceutical generic drug selection	2023
4.	Why Big Data for Pharmaceutical Product Selection: A Review	2024

Limitations

- **Limited Data Availability -** The other factors like Value , Volume, Cost Of Goods Sold(COGS),Average Selling Prize (ASP),Return On Investment(ROI),Gross Margin, Sales Data is not considered by this model because of unavailability of this data online.

Future Research

- The model was unable to collect features related to value, volume, Cost of Goods Sold (COGS), Average Selling Price (ASP), Return on Investment (ROI), Gross Margin, and Sales Data. This is because such information is confidential within the pharmaceutical industry and is not typically made available on public platforms. However, if pharmaceutical companies had access to this data, it could significantly enhance decision-making for generic drugs.
- Beyond the pharmaceutical sector, this prediction model can prove to be highly valuable for a wide range of industries seeking to make data-driven decisions regarding product selection. By leveraging the model's capabilities, businesses in sectors such as retail, manufacturing, and consumer goods can gain valuable insights into market trends, consumer preferences, and product performance, ultimately optimizing their product strategies and improving overall decision-making processes.

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Thank you